



AWS Hands-on Exploration: EC2 and S3

Summary

This document summarizes the steps, technical insights, and key concepts learned during the hands-on exploration of Amazon EC2 (Virtual Machines) and Amazon S3 (Simple Storage Service).

Amazon EC2: Deploying a High-Availability-Ready Web Server

This section covers the setup and deployment of a web server on an EC2 instance.

Service: Amazon EC2 (Virtual Machines).

Cloud Object Storage x Launch an instance x Untitled document x Verify your identity x Amazon Web Services x Google Gemini x New tab x Ask Gemini

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:tutorialId=launch-a-test-instance

aws [Option+S] Europe (Stockholm) Aditya Shankar (5891-5945-8470) Aditya%20Shankar

EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough Do not show me this message again.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name
e.g. My Web Server

Application and OS Images (Amazon Machine Images)

An AMI contains the operating system, application server, and application software. To launch an instance, you must select an AMI. To find an AMI, use the search field or choose Browse more AMIs.

Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat Enterprise Linux

Summary

Number of instances: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...read more
ami-0aaa636894689fa47

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

Task 1: Launch a test instance

Name your instance so that you can easily keep track of it. The instance name is a tag, where the key is Name, and the value is the name that you specify.

Tip: For test instances, it's okay to only specify a name tag. However, when you launch production instances, it's best practice to establish a tagging policy to standardize tagging across all your resources.

For this tutorial: Enter Test Instance in the Name field, and then choose Next.

Learn more

Step 1/8 Next

Task 1: Launch a test instance

Total steps: 8

Step 1: Name your instance
Step 2: Select an AMI
Step 3: Select an instance type
Step 4: Create a new key pair
Step 5: Edit your security group rules
Step 6: Security group settings
Step 7: Storage
Step 8: Review and launch your instance

Task 2: Post Launch Setup

Total steps: 1

Task 3: Terminate (delete) your instance

Total steps: 2

Dismiss tutorial

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Cloud Object Storage x Launch an instance x Untitled document x Verify your identity x Amazon Web Services x Google Gemini x Copy of Untitled x Ask Gemini

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:

aws [Option+S] shank's-first-aws-key.pem 387 B · Done 5891-5945-8470 Aditya%20Shankar

EC2 > Instances > Launch an instance

Access

Successfully initiated launch of instance (i-0432d495bf873f275)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your Instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 Instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

Task 2: Terminate (delete) your instance

View your instance by clicking on the instance id link, which will take you to the instance details page.

Step 1/2 Previous Next

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Action: Created an ED25519 Key Pair for secure SSH access.

Key Setup and Security Steps

Service	Action	Technical Insight
Amazon EC2	Created an ED25519 Key Pair.	Used for secure SSH access.
SSH Key	Applied <code>chmod 400</code> to the private key.	Ensures compliance with the Principle of Least Privilege (Prevents SSH failure if the key is world-readable).
Security Groups	Configured as a "Stateful Firewall."	Allowed Port 80 for public web traffic and restricted Port 22 (SSH) for controlled access.

The command used for secure connection:

```
None
ssh -i "your-key.pem" ec2-user@<Your-Public-IP-Address>
```

Web Server Deployment Details

The following software packages were installed on the Amazon Linux 2023 instance:

- `httpd-2.4.66` (Apache Web Server)
- `mod_http2-2.0.27`
- `apr-1.7.5` and related utilities (`apr-util-1.6.3`)
- Other dependencies for system operation.

EC2 Instance ConnectSSM Session ManagerSSH clientEC2 serial console

Instance ID
i-0432d495bf873f275

Connection type

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address
16.171.197.12

☐ IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.
ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

CancelConnect

Service Management

Command	Purpose
<code>sudo systemctl start httpd</code>	Starts the Apache web service.
<code>sudo systemctl enable httpd</code>	Configures the service to start automatically on boot.

Custom Web Page Creation

The default web page (`/var/www/html/index.html`) was customized to display the Instance ID, demonstrating the use of Instance Metadata.

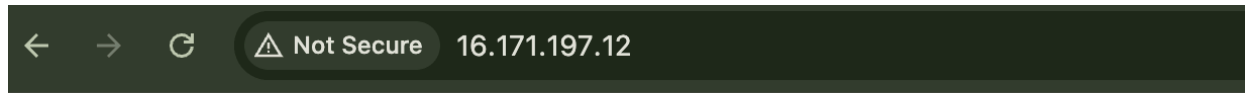
```
echo "<h1>Cloud Engineer Lab: <span type='placeholder'
placeholder-type='person'></span></h1><p>Instance ID: $(curl -s
http://169.254.169.254/latest/meta-data/instance-id)</p>" | sudo tee
/var/www/html/index.html
```

```
[aws] [x] [Q Search] [Option+S] Ask Amazon Q [x] [x] [x] [x] Europe (Stockholm) Aditya Shankar (S591-S94-S-E470) Aditya%20Shankar
```

```
Running transaction
Preparing :
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64 1/1
Installing : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 1/13
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 2/13
Installing : apr-util-1.6.3-1.amzn2023.0.2.x86_64 3/13
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 4/13
Installing : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 5/13
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 6/13
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Installing : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 9/13
Installing : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 10/13
Installing : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
Verifying : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 3/13
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.66-1.amzn2023.0.1.x86_64 6/13
Verifying : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 7/13
Verifying : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13
Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.3-1.amzn2023.0.2.x86_64      apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64    httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64     mailcap-2.1.49-3.amzn2023.0.3.noarch        mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-46-84 ~]$ sudo systemctl start httpd
sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-46-84 ~]$
```



Cloud Engineer Lab: Aditya Shankar

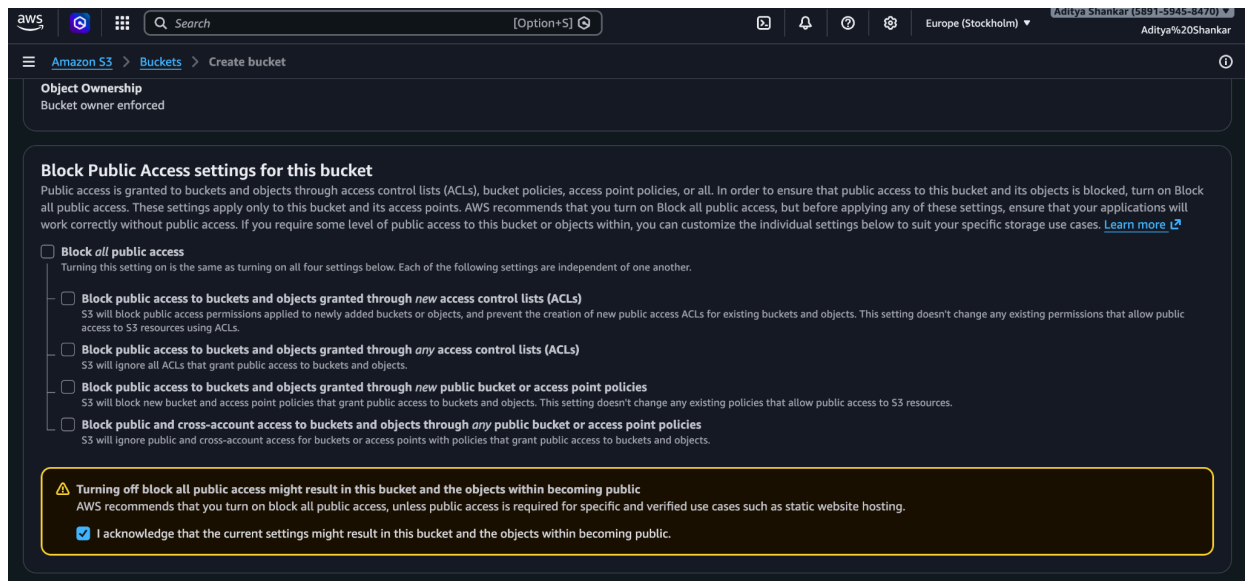
Instance ID:

Concepts Learned from EC2 Task

- **Package Management:** Used `dnf`, the package manager for Amazon Linux 2023 (successor to `yum`).
- **Systemd:** Used `systemctl` for managing services (start, enable).
- **Security Groups:** Understanding their function as a stateful, instance-level firewall.
- **Instance Metadata Service (IMDS):** Used the local service (`169.254.169.254`) to fetch runtime instance information (Instance ID).

Amazon S3: Public Content Hosting

This section covers the steps for creating and configuring an S3 bucket for public access.

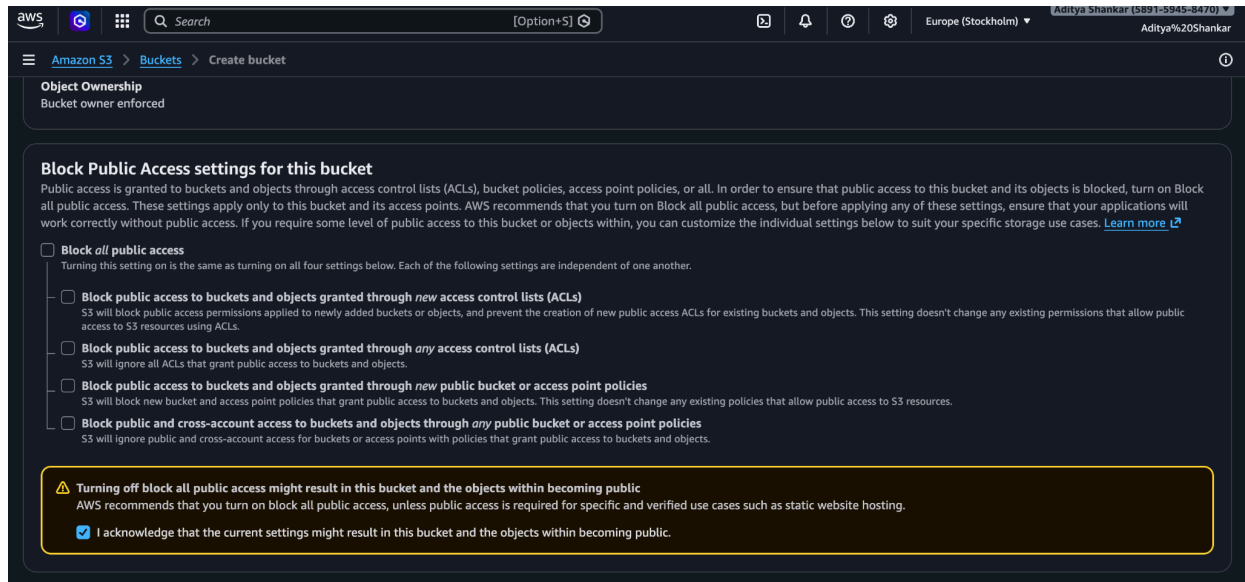


Bucket Configuration and Access

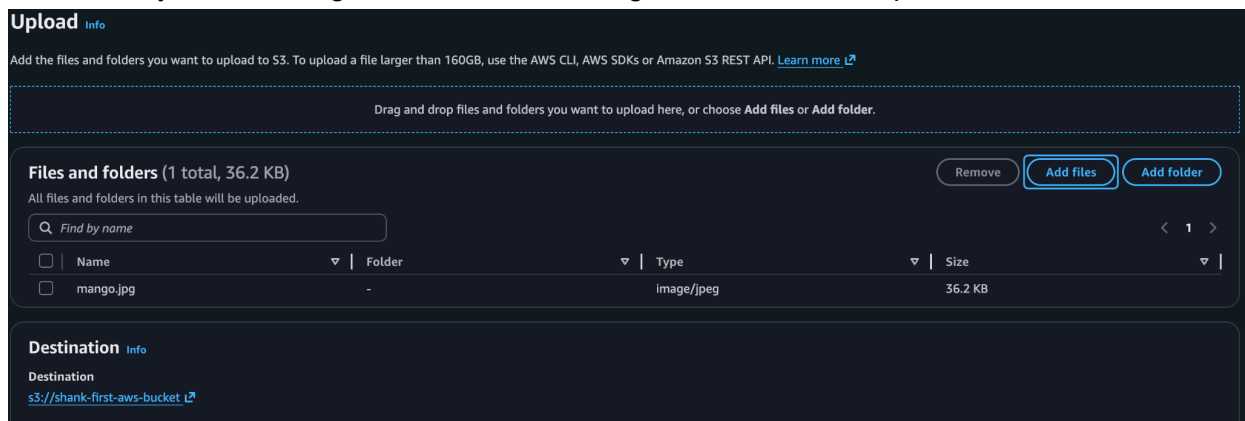
1. **Creation:** The bucket was created with the default **Block Public Access** settings *unchecked* to allow public visibility of hosted content.

2. **Upload:** Content (e.g., an image) was uploaded using the GUI.
3. **Policy Edit:** Unchecking the public access box is insufficient; a **Bucket Policy** must be edited to explicitly grant public read access.
4. **Verification:** The public URL of the stored file was copied and pasted into a new tab to confirm visibility.

Buckets :



Keep it public (uncheck the default boxes) because in this example you want to show the world the content you host using an URL. After creating, make use of the upload button in the GUI.



Simply unchecking is not enough, you'd need to edit the bucket policy too.

Amazon S3

Upload succeeded

For more information, see the [Files and folders](#) table.

Upload: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://shank-first-aws-bucket

Succeeded

1 file, 36.2 KB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 36.2 KB)

Find by name

Name	Folder	Type	Size	Status	Error
mango.jpg	-	image/jpeg	36.2 KB	Succeeded	-

aws

Search

[Option+S]

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Amazon S3

Buckets

shank-first-aws-bucket

Successfully edited bucket policy.

Bucket policy

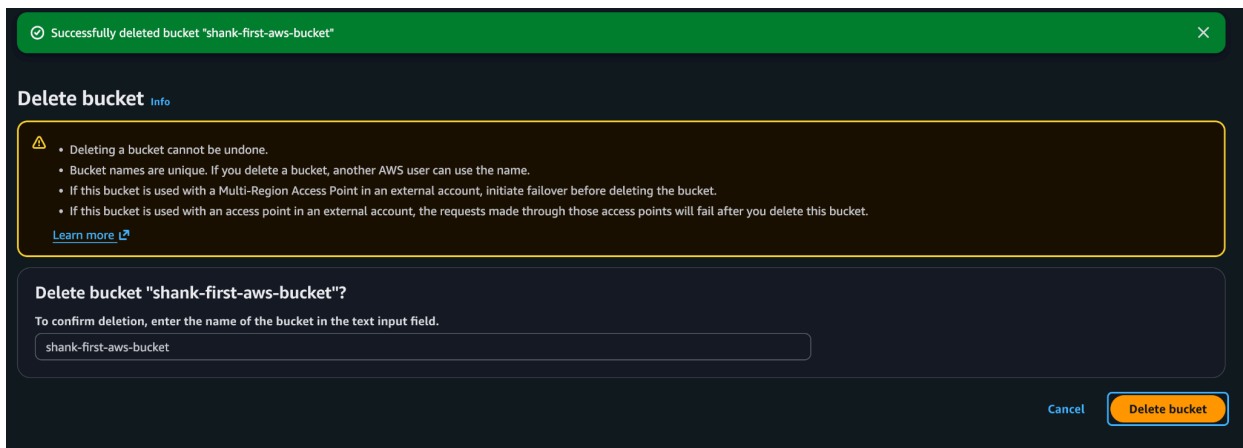
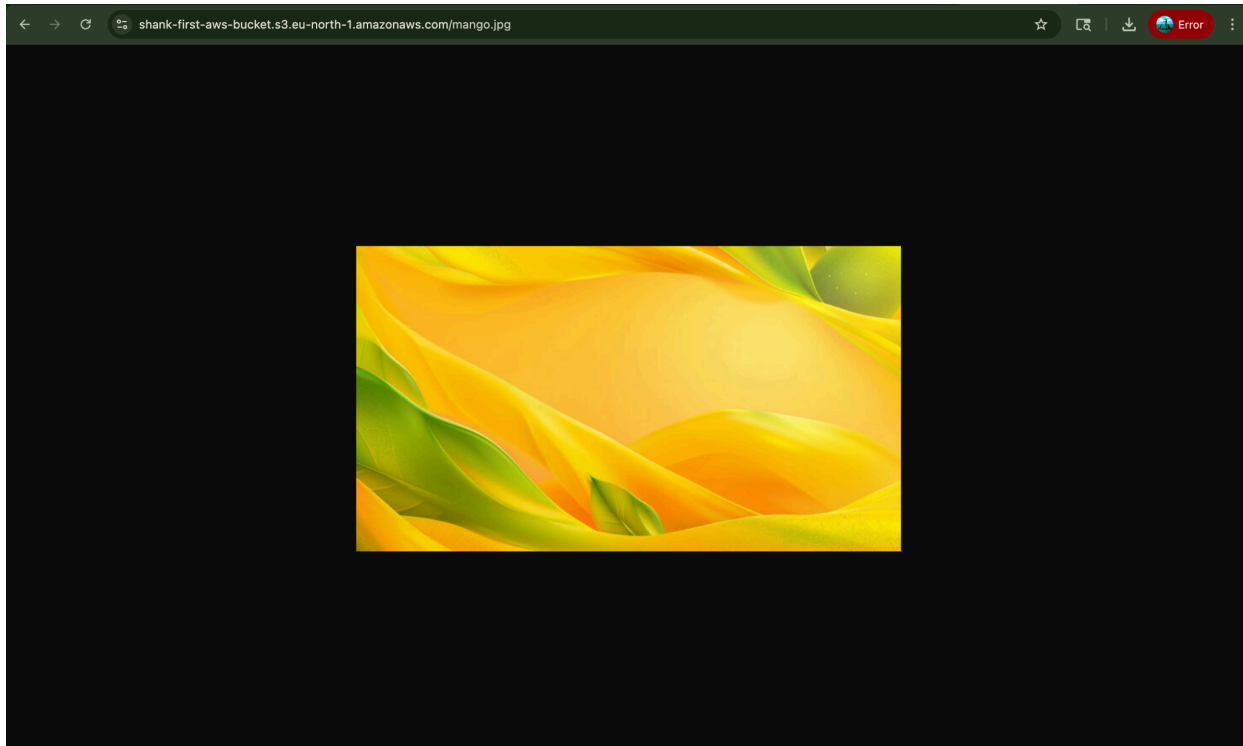
Edit

Delete

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::shank-first-aws-bucket/*"
    }
  ]
}
```

Copy



Cleanup

- The EC2 instance was stopped.
- The S3 bucket was emptied and then deleted.

Next to try:

Cloud watch

aws

Search

[Option+S]

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CloudWatch

Overview info

1h3h12h1d1wCustomUTC timezone

OverviewFilter by resource groupinfoActions



Set alarms on any of your metrics to receive notification when your metric crosses your specified threshold.
[Create alarms](#)



Create and name any CloudWatch dashboard **CloudWatch-Default** to display it here.
[Create a default dashboard](#)



Monitor using your existing system, application and custom log files.
[View logs](#)



Write rules to indicate which events are of interest to your application and what automated action to take.
[View events](#)

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Configure Application Insights

aws

Search

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CloudWatchDashboardsShankDash

ShankDash

Add widget

Data sources types

☒ Cloudwatch

☐ Other content types

☐ Create data sources

Widget Configuration

Data type

MetricsLogsAlarms

Widget type

☒ Line

Compare metrics over time



☐ Data table

Compare metrics values over time in a table



☐ Number

Instantly see the latest value for a metric

75%



☐ Gauge

See the latest value of a metric within a range



☐ Stacked area

Compare the total over time



☐ Bar

Compare categories of data



☐ Pie

Show percentage or proportional data



☐ Explorer

A single widget with multiple tag-based graphs



CancelNext

Autosave: Off

ActionsSave+